

## RESEARCH ARTICLE

# Foraging traits modulate stingless bee community disassembly under forest loss

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**Abstract**

1. Anthropogenic land use change is an important driver of impacts to biological communities and the ecosystem services they provide. Pollination is one ecosystem service that may be threatened by community disassembly. Relatively little is known about changes in bee community composition in the tropics, where pollination limitation is most severe and land use change is rapid. Understanding how anthropogenic changes alter community composition and functioning has been hampered by high variability in responses of individual species. Trait-based approaches, however, are emerging as a potential method for understanding responses of ecologically similar species to global change.
2. We studied how communities of tropical, eusocial stingless bees (Apidae: Meliponini) disassemble when forest is lost. These bees are vital tropical pollinators that exhibit high trait diversity, but are under considerable threat from human activities.
3. We compared functional traits of stingless bee species found in pastures surrounded by differing amounts of forest in an extensively deforested landscape in southern Costa Rica.
4. Our results suggest that foraging traits modulate competitive interactions that underlie community disassembly patterns. In contrast to both theoretical predictions and temperate bee communities, we found that stingless bee species with the widest diet breadths were less likely to persist in sites with less forest. These wide-diet-breadth species also tend to be solitary foragers, and are competitively subordinate to group-foraging stingless bee species. Thus, displacement by dominant, group-foraging species may make subordinate species more dependent on the larger or more diversified resource pool that natural habitats offer. We also found that traits that may reduce reliance on trees—nesting in the ground or inside nests of other species—correlated with persistence in highly deforested landscapes.
5. The functional trait perspective we employed enabled capturing community processes in analyses and suggests that land use change may disassemble bee communities via different mechanisms in temperate and tropical areas. Our results

further suggest that community processes, such as competition, can be important regulators of community disassembly under land use change. A better understanding of community disassembly processes is critical for conserving and restoring pollinator communities and the ecosystem services and functions they provide.

#### KEYWORDS

biodiversity, community disassembly, fourth-corner problem, functional traits, land use change, Meliponini, pollination, stingless bees

## 1 | INTRODUCTION

Human activities are major drivers of landscape change, leading to habitat loss, fragmentation and alteration of habitat composition. These landscape changes in turn can lead to disassembly of biological communities, i.e., non-random species declines and losses (e.g. Figuerola, Guerrero, Türkowsky, Wall, & Erijman, 2015; Leavitt & Fitzgerald, 2013; Moser et al., 2015). Understanding which taxa are better able to persist in disturbed environments and the mechanisms underlying disassembly can help guide taxon-specific conservation efforts. This understanding is necessary to predict how community disassembly will alter or reduce ecosystem services (Cardinale et al., 2012; Zavaleta et al., 2009).

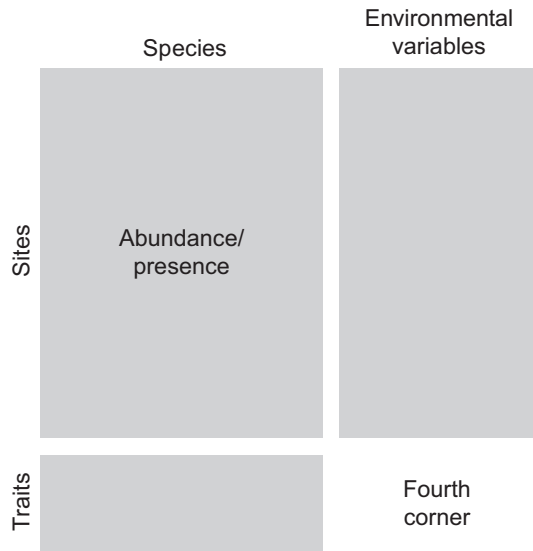
One ecosystem service globally threatened by community disassembly is pollination. Bees are considered the most important animal pollinators for most plants and ecosystems (Klein et al., 2007; Ollerton, Winfree, & Tarrant, 2011). Habitat loss due to human activities reduces bee abundance and diversity (Potts et al., 2010) and lowers plant reproductive success (Albrecht, Schmid, Hautier, & Müller, 2012; Hoehn, Tschardtke, Tylianakis, & Steffan-Dewenter, 2008). Land use change is especially rapid in the tropics, marked by both increasing cropland (Beddow, Pardey, Koo, & Wood, 2010) and decreasing forest area (Hansen et al., 2013). These changes may have stronger ecosystem-level effects in tropical than in temperate areas due to more severe pollen limitation (Vamosi et al., 2006) and a higher proportion of animal-pollinated plants (Ollerton et al., 2011). Furthermore, bee community disassembly patterns may differ between temperate and tropical regions due to the high abundance and species richness of social bees in the tropics (Roubik, 1992a). It is thus critical to determine how land use change disassembles communities of tropical bees and the plants they pollinate.

Focusing on functional traits, characteristics of an organism that strongly influence its fitness (McGill, Enquist, Weiher, & Westoby, 2006; Vandewalle et al., 2010), holds great promise for determining effects of anthropogenic changes on community composition and functioning. Functional traits, which are often shared by multiple members of a community, can better predict ecosystem functioning than do purely species-based indices such as abundance and richness (Gagic et al., 2015). This is proving true for pollination, where the degree to which bees interact with plants and provide pollination services is best explained by specific traits of the bees (e.g. mouthpart size) and flowers (e.g. whether open or tubular) (Crea, Ali, & Rader, 2016; Garibaldi et al.,

2015). Functional traits also effectively predict responses of communities to global change (e.g. Marini et al., 2012; Pedley & Dolman, 2014). This may be in part because traits strongly influence how organisms interact with each other and their environments. Such interactions can be mechanisms of community disassembly (Bregman et al., 2015) and of ecosystem functioning. For example, Forrest, Thorp, Kremen, and Williams (2015) showed that reduced bee diversity on farms compared to natural areas is in part due to a lack of sufficient nesting substrate on farms for species that nest above-ground (e.g. in trees, stems or dead wood). In addition, studying land use change within a trait-based framework facilitates determining general patterns that can be applied to related taxa or separate geographic regions (Keddy, 1992). Focusing on functional traits has helped identify general disassembly patterns of temperate bee communities, such as increased losses of solitary bees with disturbance (Forrest et al., 2015; Jauker, Krauss, Jauker, & Steffan-Dewenter, 2013; Pisanty & Mandelik, 2014; Rader, Bartomeus, Tylianakis, & Laliberté, 2014).

Despite their promise, the use of functional traits to study mechanisms of community disassembly has been hampered by a methodological issue: the “fourth-corner” problem (Dray & Legendre, 2008; Legendre, Galzin, & Harmelin-Vivien, 1997). Standard analysis methods are unable to directly analyse the relationship between a species' traits and characteristics of the environment where that species is found (Legendre et al., 1997). Because traits are properties of species (Figure 1, bottom left) while environmental variables are properties of sites (Figure 1, top right), there is an inherent challenge in estimating the joint effects of traits and the environment (Figure 1, bottom right) on community metrics such as species richness.

Several approaches attempt to work around the fourth-corner problem. Early approaches involved interpreting species-environment relationships a posteriori in light of trait data (e.g. Cane, Minckley, Kervin, Roulston, & Williams, 2006) or species-trait relationships in light of environmental data (e.g. Mabry & Fraterrigo, 2009), which precludes testing mechanistic hypotheses. Another common approach is assessing the frequency of different traits or mean trait values in different environments (e.g. Moretti, De Bello, Roberts, & Potts, 2009). Summarizing trait distributions within a site often leads to low discriminatory (Verberk, van Noordwijk, & Hildrew, 2013) and predictive (Wright et al., 2006) power, and can yield data that are overly influenced by outliers or abundant species. A common approach to minimizing this unequal influence, weighting by abundance, is problematic when data include species of varying degrees of sociality and thus



**FIGURE 1** The “fourth-corner problem” refers to the difficulty of analysing joint effects of environment and trait, because the former is a property of sites and the latter a property of species. Here, each grey box represents a data matrix or table (modified from figure 1 in Walker et al., 2012)

expected group sizes (as is often the case with bees). Summarizing trait distributions also is better suited to answering questions about changes in community composition than species-level responses to environmental change: the focus of our study. Furthermore, work-arounds cannot explicitly test how trait and environmental values simultaneously alter persistence probabilities (or other performance consequences) because they do not explicitly include a trait-by-environment interaction (McGill, 2015). This makes it difficult to directly address mechanisms that determine how traits mediate community disassembly (Pedley & Dolman, 2014). In contrast, analyses that incorporate the fourth corner, such as the one presented here, permit direct tests of mechanisms. We used the method developed by Walker, Guénard, Sólomos, and Legendre (2012) to combine the three standard “corners” (Figure 1; site-by-species matrix, species-by-traits matrix, site-by-environment matrix) into a long format that allows regression of species occurrence by both traits and environmental data in the same dataset. This approach avoids the pitfalls described above by (1) including both trait and environment data for each site–species combination and (2) combining continuous and categorical trait variables. Our regression-based method, which is not restricted to any one type of regression, provides an additional benefit: flexibility. For example, it can be used to study nonlinear effects, or to test *a priori* hypotheses about combined effects of multiple traits and environmental variables (via planned contrasts).

Here, we examine whether functional traits are associated with the disassembly of stingless bee communities under land use change. We focus on social stingless bees (Hymenoptera: Apidae: Meliponini), which are thought to provide a large share of tropical insect pollination (Heard, 1999) and to suffer disproportionate abundance and richness losses from human activities (Brosi, Daily, & Ehrlich, 2007; Brosi, Daily, Shih, Oviedo, & Durán, 2008). The strong effect of habitat

loss on these social tropical species contrasts with land use change predominantly affecting solitary species in temperate areas (Forrest et al., 2015; Gonçalves, Sydney, Oliveira, & Artmann, 2014; Pisanty & Mandelik, 2014; Rader et al., 2014). In addition to their high taxonomic diversity—over 500 species (Michener, 2000)—stingless bees within and across sites show high diversity in functionally-relevant traits. For example, body size can range from a head width of 0.8 mm (Pedro & Camargo, 2009) to 5 mm (Roubik, 1992b), and colony sizes from less than 100 (Roubik, 1983a; Wille & Michener, 1973) to over 10,000 individuals (Hubbell & Johnson, 1977; Lichtenberg, Imperatriz-Fonseca, & Nieh, 2010). Stingless bees typically nest in tree cavities or the ground, although some species build nests on tree trunks or branches. All stingless bees are eusocial, but some species forage as individuals while others tend to forage in large groups of nestmates (Johnson, 1983). With such high trait diversity and critical roles as pollinators, stingless bees are a good system for studying the dynamics of system change in response to anthropogenic disturbance.

We use data on the effects of forest loss on Costa Rican stingless bee communities to ask two questions: (1) Does land use change cause trait shifts within stingless bee communities? and (2) Does competition affect which species can persist in altered habitats? It is generally held that specialized species and larger animals are more sensitive to habitat loss and prone to extinction (Ewers & Didham, 2006; McKinney, 1997). Studies of temperate bee communities to date typically support specialization effects (Bartomeus et al., 2013; Cane et al., 2006; Forrest et al., 2015; Gonçalves et al., 2014; Kleijn & Raemakers, 2008; Rader et al., 2014; Williams et al., 2010; but see Otieno et al., 2015), but show mixed support for size effects (Bartomeus et al., 2013; Jauker et al., 2013; Rader et al., 2014; Williams et al., 2010; Winfree, Griswold, & Kremen, 2007; Wray, Neame, & Elle, 2014). Too little is known about stingless bee biology and responses to land use change to formulate specific hypotheses about other traits. In terms of the second question, competition with honeybees (Paini, 2004; Roubik, Moreno, Vergara, & Wittmann, 1986) and among stingless bees (Eltz, Brühl, van der Kaars, & Linsenmair, 2002; Slaa, 2006) have been hypothesized to be key factors shaping stingless bee communities. Forest loss can reduce both nesting sites and food (flowers) available to stingless bees (Batista, Ramalho, & Soares, 2003; Roubik, 1989), potentially intensifying competition for those resources. Thus, competition may partially drive species persistence patterns under land use change.

## 2 | MATERIALS AND METHODS

### 2.1 | Bee sampling

We sampled stingless bees in the Valle de Coto Brus, Puntarenas province, southern Costa Rica, in the landscape surrounding the Las Cruces Biological Station (8°47'N, 82°57'W), near the town of San Vito. This region was covered by premontane wet tropical forest before European settlement. It now consists of interspersed pastures, coffee, small (<1 ha) mixed-agricultural plots, human settlements and small (mostly <10 ha) forest patches (see Brosi et al., 2007 for more detail). The dataset used here is identical to that described in Brosi

(2009), consisting of 35 sites in the Las Cruces landscape. All sites were in relatively open pastures (mean area c. 0.5 ha) with active grazing rotations. Sites ranged from 900 to 1,300 m in elevation above sea level and from 500 m to 13 km in geographic distance from one another, with no spatial autocorrelation in community composition (Brosi, 2009; Brosi et al., 2007). Sites were surrounded by varying degrees of tree cover.

Each site was comprised of a 20 × 20 m plot, in which we sampled all bees detected in 15-min netting samples with two field team members actively netting. We did not sample bees in conditions of high winds, fog or precipitation. Bees were captured in the order that they were seen (i.e. rare species were not sampled at the expense of a common species seen first). This sampling effort was inclusive of all species in the bee community, but we used only the stingless bees from these samples in the analyses presented here. We sampled foraging bees in the rainy seasons of 2003 and 2004 (June–September) and in the dry season of 2005 (February–May). In Costa Rica, stingless bee richness and species' abundances show little seasonality (Heithaus, 1979). Each site was sampled on 10–22 separate days over these periods. We pinned, labelled and identified stingless bees to the species level using Roubik (1992b). V. Gonzalez and I. Hinojosa, then at the University of Kansas, evaluated and corrected species determinations. Some determinations are updated here following Camargo and Pedro (2013). Specimens are housed in the Department of Environmental Sciences, Emory University. For more on the sampling, see Brosi et al. (2007, 2008) and Brosi (2009).

For each site, we determined the presence or absence of each species from netting data pooled across all sampling days. We focused on presence/absence, rather than species' abundances, because abundances can reflect a species' propensity to recruit nestmates to rich resources. Abundances thus do not consistently represent the number of nests in a region when considering multiple species. To account for uneven sampling effort across sites (10–22 days), we used sample-based rarefaction to calculate rarefied presence/absence: the probability a species was present at a site in any set of 10 sampling days.

## 2.2 | Bee traits

We selected seven life-history and behavioural traits that are relevant to stingless bees conservation and their role as pollinators, and for which we could find enough data in the literature: body size (average head width), colony size (average number of workers), diet breadth, group foraging, nesting in the ground, inquiline nesting (nesting in other species' nests) and nesting in man-made structures. The traits we selected can reasonably be assumed to impact stingless bee nesting and foraging, even though little is known about pollinator functional traits in general. The first three are continuous and the latter four categorical (yes/no). Because diet breadth data are limited, we used diet breadth data from congeneric Brazilian stingless bees (Biesmeijer & Slaa, 2006). Following Lichtenberg et al. (2010), we used descriptions of bees foraging on natural food sources (citations in Table 1) to characterize foraging strategies. These foraging strategy classifications consider numbers

of nestmates visiting the same food source rather than recruitment. Species whose colonies can forage in large groups at the same spatial location (typically the same mass-flowering tree) were categorized as group foraging, and those whose workers forage as solitary individuals at different spatial patches were categorized as solitary foraging. For the three nesting variables, species were categorized as “yes” if at least one instance of the relevant nest type has been reported and “no” otherwise. Ground-nesting species may also nest in trees. We did not include a tree-nesting variable because almost all of the species we recorded are known to nest in or under trees (Camargo & Pedro, 2013).

## 2.3 | Site-specific data

In addition to bee traits, we assessed the impact of landscape context, bee community context and flowering plant resources on stingless bee communities. To quantify landscape context, we calculated the proportion of forest around sample points at radii ranging from 10 to 1,350 m in 10–50-m increments. We classified ecosystem elements manually, by digitizing Costa Rica Airborne Research and Technology Applications (CARTA) orthorectified aerial photographs from 2003 to 2005 with 2-m resolution. Forest elements included: primary and secondary forest fragments of all sizes, single trees, charral (early secondary forest), live fences, hedgerows, non-native timber and fruit tree plantations, and non-native garden ornamentals. These data are much finer resolution than the 30 × 30 m pixel LANDSAT data previously used by Brosi (Brosi, 2009; Brosi et al., 2008). We determined the most appropriate radius to use in analyses by assessing the correlation between percentage forest cover and stingless bee richness at each distance. Because stingless bee richness was significantly related to forest cover up to c. 250 m (Figure S1), further analyses quantify landscape context as the percentage of forest within a 250-m radius circle around each sample point.

To quantify the bee community context as an explanatory variable for our analyses of competition, we used two approaches. First, we calculated the abundance and richness of all bees sampled at each site. Second, we separately calculated abundance and richness values for stingless bees, honeybees (abundance only) and all other bees. Because sampling effort was uneven across sites, we applied sample-based rarefaction to estimate bee abundance and richness at each site if they had been sampled only 10 times (the fewest times any one site was sampled) (Gotelli & Colwell, 2011).

To quantify flowering plant resources available to stingless bees at each pasture site, we counted flowering plants along five parallel 20-m transects in each site. We counted and identified all plants in flower within 50 cm of either side of the transect line. See Brosi et al. (2007, 2008) for more details on plant sampling.

## 2.4 | Data analysis

We tested effects of trait values, landscape context, bee community context and flowering plant richness on the presence or absence of

**TABLE 1** Trait values for Costa Rican stingless bee species used in this study

| Species                                          | Name in Brosi (2009) <sup>A</sup>                 | Head width (mm) <sup>B</sup> | Colony size (# workers) <sup>C</sup> | Diet breadth <sup>2</sup> | Group foraging? <sup>D</sup> | Nests in ground? <sup>E</sup> | Inquiline nesting? <sup>E</sup> | Nests in man-made structures? <sup>E</sup> |
|--------------------------------------------------|---------------------------------------------------|------------------------------|--------------------------------------|---------------------------|------------------------------|-------------------------------|---------------------------------|--------------------------------------------|
| <i>Aparatrigona isopterothila</i>                | <i>Paratrigona isopterothila</i>                  | 2.1 <sup>7j</sup>            | 185 <sup>h</sup>                     | 0.56                      |                              | No                            | Yes                             | No                                         |
| <i>Melipona costaricensis</i> or <i>panamica</i> | <i>Melipona fasciata</i> <sup>cc</sup>            | 4.01 <sup>i,p,ee</sup>       | 1092.84 <sup>b,h,i,n,o,dd</sup>      | 0.44                      | No <sup>k</sup>              | No                            | No                              | No                                         |
| <i>Oxytrigona mellicolor</i>                     | <i>Oxytrigona</i> sp. <sup>iw</sup>               | 2.7 <sup>i</sup>             | 5442 <sup>h</sup>                    |                           | Yes <sup>u</sup>             | No                            | No                              | No                                         |
| <i>Paratrigona ornaticeps</i>                    | <i>Paratrigona ornaticeps</i>                     | 1.795 <sup>il</sup>          | 1710 <sup>h</sup>                    | 0.56                      |                              | No                            | No                              | No                                         |
| <i>Partamona orizabaensis</i>                    | <i>Partamona orizabaensis</i>                     | 2.46 <sup>s</sup>            | 2950 <sup>b,c</sup>                  | 0.37                      | Yes <sup>g,k</sup>           | Yes                           | Yes                             | Yes                                        |
| <i>Plebeia frontalis</i>                         | <i>Plebeia frontalis</i>                          | 1.8 <sup>lee</sup>           | 1500 <sup>b,c,f,h</sup>              | 0.51                      | No <sup>c,g</sup>            | No                            | No                              | Yes                                        |
| <i>Plebeia jatifformis</i>                       | <i>Noguerapis</i> sp., <i>Plebeia jatifformis</i> | 1.5 <sup>j</sup>             | 805 <sup>h</sup>                     | 0.51                      | No <sup>k</sup>              | No                            | No                              | No                                         |
| <i>Ptilotrigona occidentalis</i>                 | <i>Trigona</i> c.f. <i>ferricauda</i>             | 3.055 <sup>iv</sup>          |                                      |                           |                              | No                            | Yes                             | No                                         |
| <i>Scaptotrigona pectoralis</i>                  | <i>Scaptotrigona</i> c.f. <i>pectoralis</i>       | 2.5 <sup>i</sup>             | 4375 <sup>c,f</sup>                  | 0.26                      | Yes <sup>c,d,g,q,t</sup>     | No                            | No                              | Yes                                        |
| <i>Scaptotrigona subobscuripennis</i>            | <i>Scaptotrigona mexicana</i>                     | 2.6 <sup>j</sup>             |                                      | 0.26                      | Yes <sup>c,d,g,m,r,taa</sup> | Yes                           | No                              | No                                         |
| <i>Tetragona perangulata</i>                     | <i>Trigona</i> sp. 2, <i>Trigona perangulata</i>  | 2.7 <sup>j</sup>             |                                      | 0.53                      |                              | No                            | No                              | No                                         |
| <i>Tetragona zieglerei</i>                       | <i>Trigona dorsalis</i>                           | 2.3 <sup>j</sup>             | 2647.75 <sup>c,f,t</sup>             | 0.53                      | No <sup>c</sup>              | Yes                           | No                              | Yes                                        |
| <i>Tetragonisca angustula</i>                    | <i>Trigona angustula</i> <sup>w,bb</sup>          | 1.8 <sup>i</sup>             | 2715 <sup>a,h,n,x,y</sup>            | 0.5                       | No <sup>g</sup>              | Yes                           | No                              | Yes                                        |
| <i>Tetragonisca buchwaldi</i>                    | <i>Trigona</i> sp. 1 <sup>i</sup>                 | 1.9 <sup>j</sup>             | 1376.25 <sup>b</sup>                 | 0.5                       |                              | Yes                           | Yes                             | No                                         |
| <i>Trigona corvina</i>                           | <i>Trigona corvine</i>                            | 2.5 <sup>j</sup>             | 13,625 <sup>b,c,h,y</sup>            | 0.43                      | Yes <sup>d,e,t</sup>         | Yes                           | No                              | No                                         |
| <i>Trigona fulviventris</i>                      | <i>Trigona fulviventris</i>                       | 2.7 <sup>j</sup>             | 9375 <sup>c,f,h</sup>                | 0.43                      | Yes <sup>g,t</sup>           | Yes                           | Yes                             | Yes                                        |
| <i>Trigona silvestriana</i>                      | <i>Trigona amalthaea</i>                          | 3.3 <sup>eee</sup>           | 7500 <sup>c,f,h</sup>                | 0.43                      | Yes <sup>c,d,e,g</sup>       | No                            | No                              | No                                         |
| <i>Trigonisca</i> sp.                            | <i>Trigonisca</i> sp.                             | 1.2 <sup>i</sup>             |                                      |                           |                              | No                            | No                              | No                                         |

<sup>A</sup>Stingless bee taxonomy and systematics have recently undergone revisions. Name changes were made following the Moure bee catalogue (Camargo & Pedro, 2013) unless otherwise indicated.

<sup>B</sup>Head width indicates body size. We based *Trigonisca* body size on *Trigonisca atomaria*, the only *Trigonisca* species with published head width. Measurement of several sampled *Trigonisca* confirmed a close match with the *T. atomaria* measure.

<sup>C</sup>Colony size represents a species' average.

<sup>D</sup>We used a functional definition of group foraging that considers numbers of nestmates visiting the same food source, and defines group foraging species as those whose colonies can forage in large groups at the same spatial location (Lichtenberg et al., 2010).

<sup>E</sup>Nest characteristics determined by searching the Moure bee catalogue (Camargo & Pedro, 2013), with a species defaulting to "no" if it has not been reported to nest in the ground, in other species' nests or in man-made structures. We could not find data for *Scaptotrigona subobscuripennis*, but classified it as a group forager since all five congeners that have been classified for this variable are group foragers.

Sources: a. Lindauer and Kerr (1960); b. Wille and Michener (1973); c. Johnson (1974); d. Johnson (1983); e. Johnson and Hubbell (1974); f. Hubbell and Johnson (1977); g. Hubbell and Johnson (1978); h. Roubik (1983b); i. Roubik (1992b); j. Roubik and Buchmann (1984); k. Howard (1985); l. Camargo and Moure (1994); m. Kerr (1994); n. Biesmeijer (1997); o. Nieh and Roubik (1998); p. Biesmeijer, Smeets, Richter, and Sommeijer (1999); q. Jarau, Hrnčir, Zucchi, and Barth (2000); r. Jarau, Hrnčir, Schmidt, Zucchi, and Barth (2003); s. Pedro and Camargo (2003); t. Slaa (2003); u. Slaa, Wassenberg, and Biesmeijer (2003); v. Camargo and Pedro (2004); w. Camargo and Pedro (2013); x. Tóth, Queller, Dollin, and Strassmann (2004); y. Aguilar, Fonseca, and Camargo (2005); z. Biesmeijer and Slaa (2006); aa. Menezes, da Silva, Singer, and Kerr (2007); bb. Rasmussen and Cameron (2010); cc. JMF Camargo, pers. comm.; dd. M Eckles, pers. comm.; ee. See Table S2.

each stingless bee species using logistic regression and the R statistical system v. 3.1.3. (R Core Team, 2014). Scripts and data are available online (Lichtenberg, Mendenhall, & Brosi, 2017). We used presence or absence of each species at each site (as described above) as the response variable.

In order to include interactions between traits, which describe bee species, and landscape context, which describes sites, we combined data on sites and species into a data list (Walker et al., 2012). This data structure organizes multiple table data (here, site-by-species matrix, species-by-traits matrix, site-by-environment matrix) into a long format with one row per site-species combination, and all site and trait variables as separate columns. This data list can be used by standard R regression functions. It thus overcomes the “fourth-corner problem,” the difficulty in linking species’ traits to environmental variables (Legendre et al., 1997). This yields nested data, since environmental variables are properties of the site and traits are species-specific. However, the tight correlations between our fixed effects and potential random effects (site, species) would violate the assumption that fixed and random effects are orthogonal (Wooldridge, 2010). We thus used fixed effects models to more accurately measure relationships between traits, site characteristics and a species’ presence (Townsend, Buckley, Harada, & Scott, 2013).

To identify potential drivers of stingless bee presence in our sites, we used a four-step information theoretic model selection approach. First, we ran logistic regressions (in a generalized linear modelling framework) relating the presence/absence of stingless bees at particular sites to both trait and environmental variables, with rigorous assessment for collinearity and overdispersion (relative to a binomial distribution). We assessed collinearity using regressions that included all environmental and trait variables as main effects, but no interaction terms. These independent variables were: flowering plant richness, abundance of the bee community (one or three variables), richness of the bee community (one or two variables), forest cover as landscape context, body size, colony size, diet breadth, group foraging, ground nesting, inquiline nesting and nesting in man-made structures. Regressions with rarefied stingless bee presence/absence data (presented here), which were proportions, included sampling effort (10 sites) as a weight, while regressions with unrarefied presence/absence data (see R scripts) included the number of times a site was sampled. Models with both classifications of bee community context (all bees together vs. bees split into groups) showed evidence of collinearity (high variance inflation factors; see R scripts). Collinearity was no longer present when we removed colony size, which had strong relationships with diet breadth (Spearman’s rank correlation:  $r = -.70$ ,  $S = 618.46$ ,  $p = .008$ ) and group foraging (Mann–Whitney  $U$  test:  $U = 0$ ,  $N_1 = 14$ ,  $N_2 = 12$ ,  $p = .004$ ). Further analyses thus excluded colony size. We next reran models with the terms described above plus the interaction between each trait and forest cover. Models with the bee community split into groups additionally included the interaction between honeybee abundance and the group foraging trait, since honeybees may be in stronger competition with group-foraging than solitary-foraging species. Regressions that included interaction terms but excluded colony size showed overdispersion. We thus used the quasibinomial link in regressions, and assessed model fit via quasi-AIC,

since quasi-distributions do not allow for the strict calculation of likelihood that is needed for classic AIC calculations. Because these models included multiple parameters, we used the second-order quasi-Akaike information criterion (QAICc) for smaller sample sizes (Bartón, 2013; Burnham & Anderson, 1998).

The second step was to determine the best classification of bee community context (all bees together vs. split into groups) (Table S1). Splitting the bee community into three categories yielded the best fit (QAICc smaller than other models by at least 2.0) with unrarefied presence/absence data (Table S1). It also enabled us to investigate potential competitive effects of honeybees on group- vs. solitary-foraging stingless bee species. Thus, further analyses are based on the model with the bee community split into groups.

The third step, after establishing which richness and abundance estimates to use, was to use QAICc-based model selection to remove terms from the full model using the MuMIn package (Bartón, 2013). To control for sampling effort all potential models included stingless bee abundance, and models with unrarefied presence/absence data included the number of times a site was sampled (Gotelli & Colwell, 2011).

Fourth, because the second step yielded multiple “best” models (QAICc within 2.0 of the smallest QAICc value) and these best models all had relatively low weights (Grueber, Nakagawa, Laws, & Jamieson, 2011), we used model averaging to obtain one summary model. To assess effects of forest cover, plant richness, the bee community and stingless bee traits, we tested whether each averaged coefficient was significantly different from zero. We estimated coefficients using a “full” average with shrinkage, which assumes each term is included in every model but in some the coefficient is set to zero. This avoids biasing coefficient estimates away from zero (Burnham & Anderson, 1998) and facilitates out-of-sample prediction (Hooten & Hobbs, 2015).

## 3 | RESULTS

### 3.1 | Bee traits overview

Table 1 and Table S2 shows species’ trait values. Head widths ranged from 1.2 to 4.01 mm (mean 2.39 mm), colony sizes from 185 to 13,625 workers (mean 3,950) and diet breadth from 0.26 to 0.56 (mean 0.45). Seven of the 12 species for which we could find information are group foragers. Of the 18 species studied, seven nest in the ground, five are nesting inquilines and six nest in man-made structures.

### 3.2 | Trait shifts within stingless bee communities

Results from analyses with rarefied and unrarefied presence/absence data were qualitatively similar. We thus focus on rarefied presence/absence results here, and show unrarefied presence/absence results in published R scripts (Lichtenberg et al., 2017). The averaged model included the number of times a site was sampled, plant richness, abundance and richness of the three bee categories (honeybees, stingless bees, other bees), forest cover, all six bee traits (excluding colony size, as described in the Methods), interactions between forest cover and

all bee traits, and the interaction between honeybee abundance and group foraging (Table 2).

As stingless bee communities disassembled, effects of forest loss depended on a species' diet breadth (Table 2). Species with narrower diet breadths were less likely to be present at sites with more forest nearby, while species with broader diets were more likely to be present at such sites (Figure 2a, Figure S2). Further investigation showed that diet breadth was narrower for group-foraging species than for those that are not known to forage in large groups (Figure 2b; Mann-Whitney  $U$  test:  $U = 30$ ,  $N_1 = 5$ ,  $N_2 = 6$ ,  $p = .007$ ).

The averaged model also indicated several traits that affected the probability that a species is present at a site, independent of forest availability (Table 2). Species with larger body size were more likely to be found in pastures (Figure 3a). In addition, ground-nesting (Figure 3b) and inquiline-nesting (Figure 3c) species were more common in pastures relative to species with other nesting habits.

### 3.3 | Competition and stingless bee persistence

Competition with other bee taxa did not appear to have a strong impact on whether stingless bees were found in pastures. Stingless bee

species' presence was not affected by the richness or abundance of honeybees or other bees, or by stingless bee abundance (Table 2). While a given species was more likely to be found in samples if the site had higher stingless bee richness (Table 2), this is likely to be a sampling artefact rather than reflecting facilitation among stingless bees. Furthermore, plant richness (food availability) in the sampled pastures did not affect a stingless bee species' presence in that pasture (Table 2).

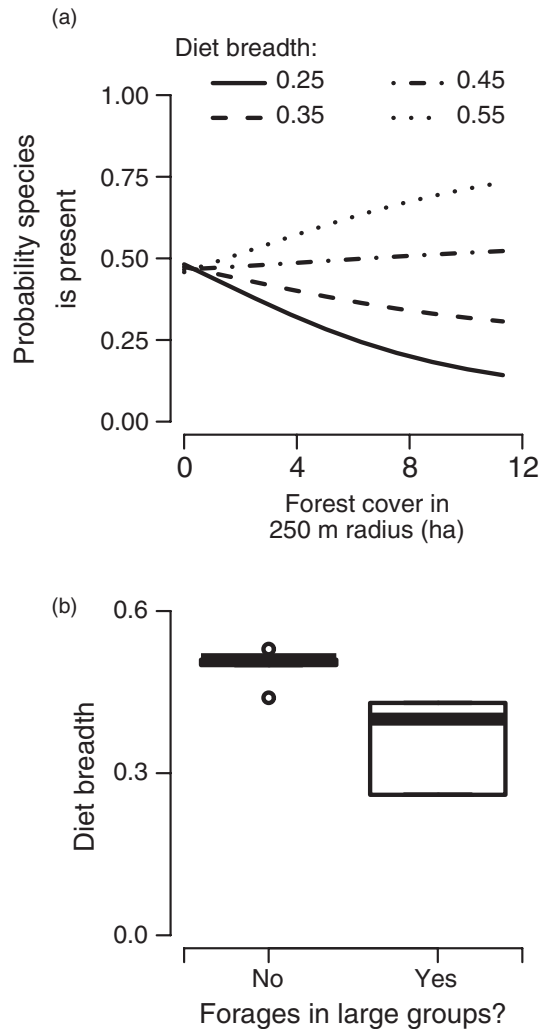
## 4 | DISCUSSION

It is increasingly clear that functional traits underlie general patterns of community disassembly with anthropogenic change. Such predictions will be useful for conservation of bees and the pollination ecosystem service they provide, but the geographic bias in bee biodiversity research (Archer, Pirk, Carvalheiro, & Nicolson, 2014) may affect assessment of community disassembly patterns. The results of this study show that social tropical bees respond to land use change differently than do temperate bees. In contrast to the majority of studies to date (e.g. Cane et al., 2006; Williams et al., 2010), we found

**TABLE 2** Weights, coefficients and Wald test results for the averaged model. Terms in bold had a statistically significant effect on a species' presence in pasture sites

| Covariate                           | w                 | Predicted coefficient ( $\pm$ SE) | z           | p-value          |
|-------------------------------------|-------------------|-----------------------------------|-------------|------------------|
| Intercept                           | —                 | −7.42 (2.98)                      | 2.49        | .013             |
| Plant richness                      | 0.29              | −0.01 (0.02)                      | 0.29        | .77              |
| Honeybee abundance                  | 0.37              | −0.003 (0.01)                     | 0.21        | .84              |
| Stingless bee abundance             | 1.00 <sup>a</sup> | −0.0007 (0.007)                   | 0.10        | .92              |
| <b>Stingless bee richness</b>       | <b>1.00</b>       | <b>0.59 (0.10)</b>                | <b>5.91</b> | <b>&lt;.0001</b> |
| Other bee abundance                 | 0.38              | −0.01 (0.02)                      | 0.42        | .68              |
| Other bee richness                  | 0.33              | −0.01 (0.06)                      | 0.11        | .91              |
| Forest cover                        | 1.00              | −3.75 (7.02)                      | 0.53        | .59              |
| <b>Body size</b>                    | <b>0.94</b>       | <b>1.19 (0.56)</b>                | <b>2.12</b> | <b>.035</b>      |
| Diet breadth                        | 1.00              | −0.62 (4.15)                      | 0.15        | .88              |
| Group foraging?                     | 0.89              | 0.47 (0.81)                       | 0.58        | .56              |
| <b>Ground nesting?</b>              | <b>1.00</b>       | <b>1.78 (0.66)</b>                | <b>2.71</b> | <b>.007</b>      |
| <b>Inquiline nesting?</b>           | <b>1.00</b>       | <b>2.34 (0.94)</b>                | <b>2.50</b> | <b>.013</b>      |
| Man-made nests?                     | 0.94              | 0.08 (0.82)                       | 0.10        | .92              |
| Forest cover: body size             | 0.94              | −1.96 (1.16)                      | 1.69        | .09              |
| <b>Forest cover: diet breadth</b>   | <b>1.00</b>       | <b>22.33 (11.38)</b>              | <b>1.96</b> | <b>.049</b>      |
| Forest cover: group foraging?       | 0.89              | 1.10 (2.00)                       | 0.55        | .59              |
| Forest cover: ground nesting?       | 1.00              | −1.56 (1.57)                      | 0.99        | .32              |
| Forest cover: inquiline nesting?    | 1.00              | 1.37 (2.34)                       | 0.59        | .56              |
| Forest cover: man-made nests?       | 0.94              | −2.68 (2.07)                      | 1.30        | .20              |
| Honeybee abundance: group foraging? | 0.15              | 0.004 (0.01)                      | 0.30        | .76              |

<sup>a</sup>Included in all models to control for effects of unequal sampling on our response variable.



**FIGURE 2** (a) Predicted probability that a species was present in a pasture as a function of the site's natural habitat availability (x-axis) and the bee species' diet breadth (individual curves). Curves show the predicted probability averaged across all values of the traits other than forest cover and diet breadth. Figure S2 shows curves with 95% confidence intervals. (b) Species that forage as solitary individuals rather than in large groups of nestmates tend to have wider diets than group-foraging species

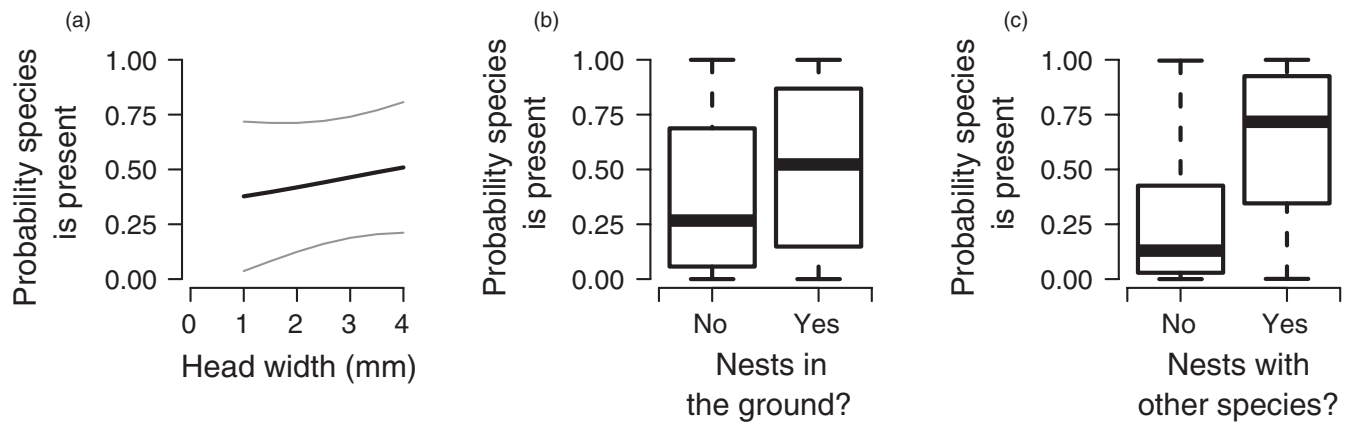
that species with narrower diets better persisted in disturbed areas. Further investigation of species' traits led us to propose the mechanistic hypothesis that foraging traits such as group foraging, and associated abilities to dominate resources, can modulate effects of land use change. Specifically, subordinate species may be frequently displaced from rich food sources and thus depend on the larger resource pool that forests offer. This analysis also found that ground and inquiline nesting, and larger body size increased the probability of finding a bee species in pastures.

Disturbance is generally predicted to reduce the abundance or diversity of more specialized species (McKinney, 1997; Vandewalle et al., 2010). Contrary to this, we found that species with the broadest diets were more sensitive to habitat loss while bee species that visit fewer plant species were largely absent in sites with high forest availability.

The only other trait-based study to date of anthropogenic effects on tropical bee communities similarly found that habitat degradation reduced abundances of polylectic bees that collect pollen from a broad range of floral sources (Otieno et al., 2015). Temperate studies, on the other hand, do follow the expected pattern (Cane et al., 2006; Forrest et al., 2015; Gonçalves et al., 2014; Kleijn & Raemakers, 2008; Rader et al., 2014; Williams et al., 2010). This contrast suggests that lower dietary specialization of pollinators in the tropics (Schleuning et al., 2012) and decreased risk of extinction in communities with more generalists (Stroud & Feeley, 2015) could leave more room for community processes to modify effects of land use change in tropical than in temperate pollinator communities. It is also worth noting that even the most "specialized" stingless bees are still generalists who visit a diverse pool of flowering plant resources, although the diversity of plant species that different stingless bee genera visit spans a factor of two (Shannon index:  $H' = 0.26\text{--}0.56$ , Biesmeijer & Slaa, 2006).

The trait-based approach we used suggests a mechanism through which forest loss reduces persistence of species with broader diets in communities of social, tropical bees: interspecific competition and dominance-related foraging traits. In this study, species with more diversified diets and a negative response to forest loss also tended to forage as individuals or in small groups. Such species are generally subordinate (Lichtenberg et al., 2010), and thus subject to displacement by group foraging, dominant species. We propose, therefore, that stingless bee species' diet breadths are plastic responses to competitive interactions rather than the evolved preferences hypothesized by earlier work (Johnson & Hubbell, 1975). This may be true even though our analyses did not detect effects of community context. Such facultative expansion of diet in response to competition is common (Futuyma & Moreno, 1988), and may help animals avoid unsustainably long food search times (Prinzinger, 2003). Under this scenario, species with broader diets may tolerate disturbance poorly if they require the larger or more diversified resource pool that natural habitats offer, or cannot compete with more dominant stingless bee species in areas with a reduced floral resource pool. The dominant species with narrower diet breadths, meanwhile, may have been largely absent from sites surrounded by more forest if those forest fragments provided sufficient quantities of one preferred food resource for stingless bees: mass-flowering trees (Ramalho, 2004). Our results thus highlight the importance of including behavioural traits in studies of community disassembly.

Our results show that ground- and inquiline-nesting stingless bee species are more likely to be found in pastures. Such species may more easily disperse through disturbed landscapes and persist in pastures than tree-nesting stingless bee species. Stingless bees establish new nests only after multiple trips between the current and the new site (Roubik, 1989), so are restricted in how far the next generation can disperse from the natal nest. Nesting in the ground, or inside the nests of taxa such as subterranean termites or mound-producing ants, may thus facilitate bee dispersal across disturbed habitat. Consistent with this hypothesis, termite mounds were common in the pastures we sampled (B. Brosi, pers. obs.). Beyond stingless bees, ground nesting encompasses many nest types and is not closely linked to a dispersal



**FIGURE 3** Predicted probability that a species was present in a pasture as a function of (a) the species' body size, as indicated by head width (95% confidence interval in grey), (b) whether the species nests in the ground and (c) whether the species exhibits inquiline nesting

mechanism. Reflecting this diversity, ground-nesting bee species have mixed responses to disturbance (e.g. Bartomeus et al., 2013; Cane et al., 2006; Forrest et al., 2015; Fortel et al., 2014; Williams et al., 2010). Species that nest above-ground in trees or stems, in contrast, are often harmed (e.g. Forrest et al., 2015; Pisanty & Mandelik, 2014; Williams et al., 2010; Winfree et al., 2007).

We also found that species with larger body size were more common in pastures. This pattern likely reflects flight range capabilities. As central place foragers, bees have a stationary nest to which they return after each foraging trip. Larger bee species are able to forage further from the nest (Araújo, Costa, Chaud-Netto, & Fowler, 2004; Greenleaf, Williams, Winfree, & Kremen, 2007), and thus large stingless bees in the present study were more likely to be seen in pastures even if their nests were in forest fragments. A stronger effect of disturbance on small than large bee species has sometimes been found in temperate studies (Williams et al., 2010; Winfree et al., 2007), although other patterns have been reported as well, including stronger effects of disturbance on larger bees (Bartomeus et al., 2013; Jauker et al., 2013; Kormann et al., 2015; Pisanty & Mandelik, 2014; Rader et al., 2014; Wray et al., 2014) and no differential effect of disturbance based on body size (Cane et al., 2006; Ekroos, Rundlöf, & Smith, 2013; Forrest et al., 2015; Fortel et al., 2014; Kleijn & Raemakers, 2008).

Trait-based approaches, as we have employed here, have some important limitations. Researchers are often restricted to traits that are already published, or that are straightforward and economical to measure in the field. Thus, trait-based analyses may overlook important ecological processes due to missing data. Analyses may also be biased towards more easily measured traits (e.g. size) or taxonomic groups (e.g. plants). Recent efforts to standardize trait measurement (e.g. Cornelissen et al., 2003; Fountain-Jones, Baker, & Jordan, 2015) and the increasing popularity of trait-based studies (Martin & Isaac, 2015) should help overcome these issues in the future. The narrow scope of our study avoids taxonomic biases, but the differences we find between effects of disturbance on bee traits in tropical and temperate regions highlights that studies assessing broader effects on all bees need to incorporate more tropical bees. Although the traits we could analyse were limited by what has

previously been published, the set of traits we used is sufficiently diversified to reasonably describe stingless bees' niches. The traits we used described species' nesting requirements, movement patterns and food needs. Data availability did limit us to assigning diet breadths at the genus rather than the species level. However, several studies find higher diet similarity within than among stingless bee genera (Imperatriz-Fonseca, Kleinert-Giovannini, & Ramalho, 1989 and sources therein; Biesmeijer & Slaa, 2006; Ramalho, Kleinert-Giovannini, & Imperatriz-Fonseca, 1990). Another limitation of trait-based approaches is subjective assessments of whether a trait is functional in the absence of data relating trait values to fitness. While we do not know the degree to which each trait affects fitness, all the traits we used have a logical connection to pollinator functioning.

One of the key challenges of the ongoing biodiversity crisis is understanding the ecosystem-level consequences of losses of populations and species (e.g. Hooper et al., 2005). This is particularly true in the case of pollinators, which are undergoing global declines (Potts et al., 2010) and which also play a key functional role in the pollination of both crops (Klein et al., 2007) and wild plants (Ollerton et al., 2011). Loss of habitat is one of the key drivers of pollinator—and more specifically bee—diversity (e.g. Potts et al., 2010). The work we present here shows that stingless bee functional traits likely mediate, at least in part, their responses to habitat loss. It is particularly important that we continue to work to better understand how pollinator communities disassemble with habitat loss, and what the resulting consequences for pollination of both crops and native plants will be. Such an understanding could provide key insights for conserving and restoring both pollinator communities and the ecosystem services and functions that they provide.

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## AUTHORS' CONTRIBUTIONS

E.M.L. and B.B. conceived the ideas and designed methodology; all authors collected the data; E.M.L. analysed the data and led the writing of the manuscript. All authors contributed critically to drafts and gave final approval for publication.

## DATA ACCESSIBILITY

Data and R scripts are archived at Zenodo <https://doi.org/10.5281/zenodo.843615> (Lichtenberg et al., 2017).

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## SUPPORTING INFORMATION

Additional Supporting Information may be found online in the supporting information tab for this article.

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